POLY PMNA - Polytechnic Study Hub
Course 6031A - Energy Conservation & Audit (EE)
Diploma in Electrical and Electronics Engineering - Semester 6
Revision 2021 | Program Elective | Credits: 4 | Periods: 60
Downloadable handbook notes generated from lessons-6031A.html
This PDF follows the uploaded official syllabus: energy scenario, conservation, BEE star labelling, energy management, cogeneration and energy audit.

Course Overview

Objectives:
- Understand global energy scenario and environmental issues.
- Apply engineering practices in energy conservation activities.
- Select suitable energy resources for sustainable development.
- Summarize salient features of Energy Conservation Act.
- Familiarize with energy auditing, data analysis and corrective actions.

Prerequisite: Power and energy from Fundamentals of Electrical & Electronics Engineering.

Course Outcomes:
CO1: Explain the need of energy conservation and BEE star labelling.
CO2: Identify energy management techniques in electrical systems.
CO3: Illustrate energy conservation through cogeneration.
CO4: Summarize energy audit in electrical systems.

Learning Roadmap

Module 1: Energy Scenario, Conservation and BEE Star Labelling.
What to explain: energy resources, sustainable resources, energy security, BEE schemes, labels and MEPS.
What to calculate: simple energy saving by star rated devices.
What to draw: appliance comparison chart and energy conservation flow.

Module 2: Energy Management in Electrical Systems.
What to explain: lighting, motors, transformers, soft starters, VFD, APFC and IPFC.
What to calculate: energy saving with efficient motors and lighting.

Module 3: Cogeneration.
What to explain: principle, need, advantages, system types, selection criteria and prime movers.
What to draw: steam turbine, gas turbine and reciprocating engine cogeneration block diagrams.

Module 4: Energy Audit.
What to explain: audit types, instruments, data sheets, ten-step methodology, report and Sankey diagram.
What to draw: Sankey diagram and audit process flow.

Module 1 - Energy Scenario and BEE Labelling

Energy resources may be conventional, non-conventional, renewable, non-renewable, commercial and non-commercial. Sustainable resources are selected with environmental impact and long-term availability in mind.
Energy conservation means reducing avoidable energy waste without reducing useful work. It improves energy security, reduces operating cost and reduces environmental burden.
Energy intensity is a definition-level topic: energy consumed per unit output or economic activity. Lower energy intensity means better efficiency.
Energy efficiency is getting the same useful output with less input energy. Benefits include lower bills, reduced losses, better equipment life and less environmental impact.
BEE schemes under Energy Conservation Act 2001 include ECBC, standards and labelling, demand side management, Bachat Lamp Yojana, SME efficiency promotion, designated consumers, and certification of energy auditors and managers.
Star labelling helps consumers compare appliance efficiency. A higher star rating means lower consumption for the same useful output. Comparative labels show relative performance. Endorsement labels certify that a product meets a required efficiency level.
MEPS means Minimum Energy Performance Standards. It prevents inefficient products from entering the market.
Exam tip: For BEE questions, write purpose, scheme list, benefit and one appliance example.

Module 2 - Energy Management Techniques

Energy management is the systematic planning, monitoring and improvement of energy use. Objectives include reducing energy cost, reducing losses, improving reliability and supporting sustainability.
Lighting opportunities include replacing inefficient lamps, using LED fixtures, proper lux level design, cleaning luminaries, using daylight and automatic controls.
Energy efficient motors reduce losses by better magnetic material, optimized design, improved cooling and better manufacturing. They are useful for long-running loads such as pumps, fans and blowers.
Motor efficiency improvement includes correct sizing, avoiding rewinding loss, using VFD for variable load, maintaining voltage balance and reducing mechanical friction.
Energy efficient transformers reduce core and copper losses. Amorphous transformers reduce no-load loss and are useful in distribution systems with long energized hours.
Energy conservation equipment includes soft starter, automatic star-delta converter, VFD, APFC and IPFC. VFD saves energy in variable torque loads by controlling motor speed.
APFC improves power factor automatically by switching capacitor banks. It reduces penalty and improves system capacity.

Module 3 - Cogeneration

Cogeneration is the combined production of electricity and useful heat from the same fuel source. It is also called combined heat and power.
Principle: heat rejected from one process is recovered and used in another process instead of wasting it.
Need: many industries require both electricity and process heat. Separate production wastes fuel; cogeneration improves overall efficiency.
Advantages include fuel saving, lower emissions, better overall efficiency, reduced energy cost and reliable local power.
Technology based systems include steam turbine, gas turbine and reciprocating engine cogeneration.
Steam turbine cogeneration may use back pressure turbine or extraction condensing turbine.
Gas turbine cogeneration uses hot exhaust for process heat or steam generation.
Reciprocating engine cogeneration uses jacket water and exhaust heat.
Selection criteria include heat-to-power ratio, load pattern, fuel availability, operating hours, capital cost, maintenance skill, reliability and environmental limits.

Module 4 - Energy Audit of Electrical Systems

Energy audit is a systematic inspection, measurement and analysis of energy use to identify saving opportunities. The EC Act definition should be written when asked.
Need: audit identifies losses, inefficient equipment, poor operating practices, billing errors and opportunities for conservation.
Types of audit: preliminary audit is quick walk-through; targeted audit studies a selected system; detailed audit uses complete measurement and analysis.
Instruments include energy meter, power analyser, clamp meter, lux meter, tachometer, thermometer/thermal imager, flow meter and data logger.
Audit methodology: form team, collect bills, prepare data sheets, conduct walk-through, measure loads, analyse consumption, identify losses, recommend actions, estimate savings and payback, prepare report, verify after implementation.
Sankey diagram represents energy input, useful output and losses with flow widths proportional to energy quantity. It helps visualise where energy is lost.

Formula and Rule Bank

Energy saving = Old consumption - New consumption.
Cost saving = Energy saving in kWh x tariff per kWh.
Annual saving = Daily saving x operating days per year.
Simple payback = Investment / Annual cost saving.
Efficiency = Useful output / Input x 100 percent.
APFC purpose: improve power factor by automatic capacitor switching.
VFD rule: energy saving is high for variable speed fan and pump loads.
Audit rule: Do not recommend corrective action without measurement or reasonable data.
Sankey rule: width of each flow should represent energy quantity.

Solved Examples

Example 1: Star rated appliance saving.
Old appliance = 2.4 kWh/day. New appliance = 1.6 kWh/day. Saving = 0.8 kWh/day. Annual saving = 0.8 x 365 = 292 kWh. At Rs.6/kWh, cost saving = Rs.1752/year.

Example 2: Simple payback.
Investment = Rs.12000. Annual saving = Rs.3000. Payback = 12000/3000 = 4 years.

Example 3: Lighting retrofit.
50 lamps of 40 W are replaced by 18 W LED lamps. Usage = 8 h/day. Power saved = 50 x 22 W = 1100 W = 1.1 kW. Daily saving = 1.1 x 8 = 8.8 kWh.
Monthly saving = 8.8 x 30 = 264 kWh.

Example 4: Efficient motor.
Old input = 12 kW. New input = 10.5 kW. Operation = 2000 h/year. Annual saving = 1.5 x 2000 = 3000 kWh.

Practical and Industrial Applications

Control panels: APFC panels, energy meters and power analysers are directly related to energy audit.
Motors and drives: check motor loading, VFD suitability, voltage unbalance and maintenance condition.
Lighting systems: check lux level, lamp type, operating hours and automatic control options.
Transformers: check loading, no-load loss, temperature rise and efficiency class.
Escalator/elevator relevance: VFD operation, motor running hours, standby loads, lighting loads and maintenance records can be audited for savings.
Safety: never connect audit instruments without proper rating, PPE, isolation plan and supervision.

Quick Revision

Energy conservation means reducing waste, not reducing useful work.
BEE star label helps compare appliance efficiency.
MEPS sets minimum energy performance.
APFC improves power factor by capacitor switching.
VFD saves energy in variable speed motor loads.
Cogeneration produces electricity and useful heat from the same fuel.
Preliminary audit is quick; detailed audit is measurement based.
Sankey diagram shows energy input, output and losses.
Energy audit report must include observations, measurements, recommendations, saving estimate and payback.

Expected Questions

One mark: Define energy intensity. What is BEE? What is MEPS? What is cogeneration? What is Sankey diagram?
Short answer: Explain need of energy conservation. List benefits of star labelling. Name instruments used in energy audit. Explain APFC.
Three mark: Explain lighting energy saving opportunities. Explain energy efficient motor features. Compare preliminary and detailed audit.
Descriptive: Explain BEE schemes under Energy Conservation Act 2001. Explain cogeneration systems. Explain ten-step energy audit methodology.
Numerical: Estimate saving after replacing lamps or appliances. Calculate payback period. Calculate annual energy saving for efficient motors.

Model Question Paper

Time: 3 Hours. Maximum Marks: 100.

Part A: Answer all questions.

1. Define energy intensity.
2. What is MEPS?
3. Name two instruments used for energy audit.
4. What is cogeneration?
5. What is the role of Sankey diagram?

Part B:

6. Explain BEE star labelling and its benefits.
7. Explain energy conservation techniques in lighting systems.
8. Explain energy efficient motors and their limitations.
9. Explain preliminary, targeted and detailed energy audits.

Part C:

10. Explain schemes of BEE under Energy Conservation Act 2001.
11. Calculate annual saving when 60 lamps of 40 W are replaced by 18 W LEDs for 10 h/day at Rs.7/kWh.
12. Explain cogeneration systems using steam turbine, gas turbine and reciprocating engine.
13. Explain energy audit methodology and report format.

Complete Model Answers

1. Energy intensity is energy consumed per unit output or economic activity.
2. MEPS means Minimum Energy Performance Standards.
3. Instruments: power analyser, clamp meter, lux meter, thermal imager, energy meter and data logger.
4. Cogeneration is combined production of electricity and useful heat from same fuel source.
5. Sankey diagram visually shows energy input, useful output and losses with proportional flow width.
6. Star labelling: Explain BEE label, star rating, comparative/endorsement label, consumer benefit and energy saving example.
7. Lighting: LED replacement, correct lux level, daylight use, occupancy sensors, timer control, maintenance and cleaning.
8. Efficient motor: lower losses, better design, high initial cost, useful for long running loads. Mention correct sizing and VFD for variable load.
9. Audit types: preliminary is walk-through, targeted focuses on selected system, detailed audit uses complete measurements and report.
11. Power saved = 60 x (40-18) = 1320 W = 1.32 kW. Daily saving = 1.32 x 10 = 13.2 kWh. Annual saving = 13.2 x 365 = 4818 kWh. Cost saving = 4818 x 7 = Rs.33726.